

Practice Quiz: Action (Neuroscientific Frontiers)

Name: __ Date: __

Part A: Multiple Choice

1. In Active Inference, movements are caused by: A) Motor commands sent from M1 to muscles B) Proprioceptive predictions fulfilled by spinal reflex arcs — M1 sends WHERE the limb should be, not HOW to move it C) Direct cortical control of individual muscles D) Random motor neuron firing that is selected by outcome
2. The cerebellum functions primarily as a: A) Memory storage center for procedural skills B) Forward model that predicts sensory consequences of motor actions before they occur C) Backup motor cortex activated during injury D) Simple timing mechanism with no predictive function
3. Climbing fiber signals from the inferior olive represent: A) Motor commands for complex movements B) Motor prediction errors — the mismatch between predicted and actual sensory consequences of actions C) Reward signals for successful movements D) Inhibitory signals that prevent movement
4. In the basal ganglia, the direct pathway (D1) implements: A) Global inhibition of all actions B) "Go" signal for the selected policy — facilitating the action with lowest expected free energy C) "No-go" for all competing actions D) Sensory prediction errors
5. You cannot tickle yourself because: A) Your skin becomes desensitized to self-touch B) Your cerebellar forward model predicts the sensory consequences, and the predicted sensation is subtracted from the actual sensation C) Self-touch activates different nerve fibers D) Your brain ignores all self-generated sensory input
6. The readiness potential (Bereitschaftspotential) is best interpreted in Active Inference as: A) The moment the brain "decides" to act B) A free will signal from the prefrontal cortex C) Gradual accumulation of precision for the upcoming motor policy — not a discrete "decision" D) An artifact of EEG recording with no functional significance

7. Schizophrenia patients can tickle themselves because: A) Their skin is more sensitive than normal B) Their forward model for self-generated actions is imprecise — self-generated sensory consequences are not properly predicted and attenuated C) They have enhanced motor control D) They lack a cerebellum

8. Saccadic eye movements are best described as: A) Random visual scanning B) Epistemic actions that direct the fovea toward regions of maximum expected information gain C) Purely reflexive responses to visual stimuli D) Movements controlled entirely by the cortex without prediction

Part B: Short Answer

1. A patient with cerebellar damage tries to reach for a glass of water and consistently overshoots the target (hypermetria). Explain this deficit in Active Inference terms. What has failed in the patient's motor system? (200 words)

2. Compare movement vigor in Parkinson's disease vs. Huntington's disease using the basal ganglia model. What specific pathway disruption explains bradykinesia (slow movements) in Parkinson's, and what explains chorea (involuntary movements) in Huntington's? Map each to the D1/D2 balance and policy precision. (300 words)

Part C: Essay

1. A neuroscience colleague argues: "The 'predictions not commands' framework is just semantics — whether you call the signal from M1 a 'command' or a 'prediction,' the outcome is the same." Write a 400-word response defending the Active Inference interpretation. Include at least three empirical findings that distinguish predictions from commands, and explain why the distinction matters for understanding motor disorders.

2. Design an experiment that could distinguish between the "predictions not commands" framework and the classical motor command framework. Your experiment should: (a) make a specific prediction that ONLY the Active Inference account makes, (b) describe the methods needed to test it, and (c) explain what result would confirm vs. falsify the Active Inference account. (400 words)